

Reading MS Excel files

ENVX2001 – Applied Statistical Methods

Semester 1

We will work with the `diatoms.xlsx` file in this section. Download the file from Canvas or your course materials.

Introduction

i Note

Starting this week you will be regularly reading data from both `.csv` and `.xlsx` files for your labs and assignments. We have found that many students struggle with this even though they have been taught how to do it in ENVX1002, so we have created this section as a refresher, focusing on the more complex `.xlsx` file format. If you already know how to import files like a pro, feel free to skip ahead.

Data import in R can be tricky because it involves learning how to work with file paths and working directories. In this section we will import the `Diatoms` worksheet from the `diatoms.xlsx` file. This data has been mixed with other information such as exploratory work and calculations, so we will need to specify the range of cells that we are interested in to import the data appropriately.

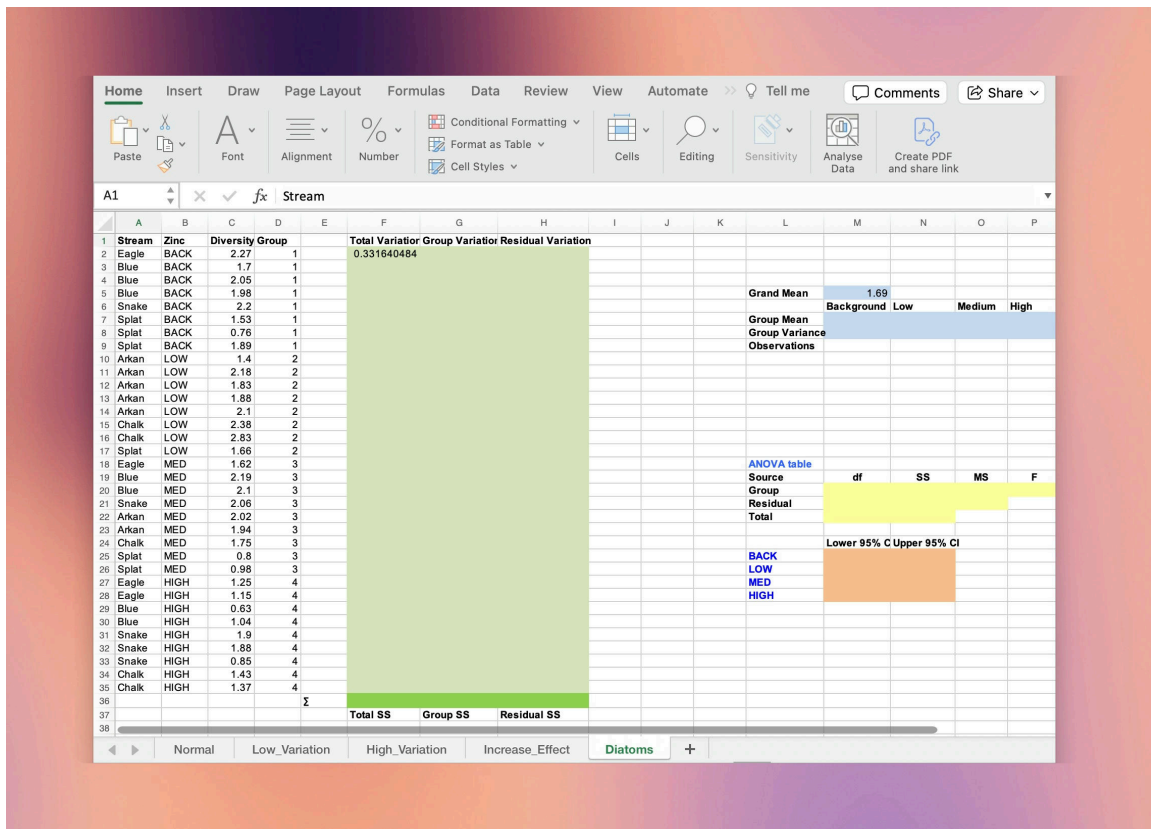


Figure 1: We need to separate the raw data (which we are interested in) from other information such as exploratory work and calculations (which we may *not* be interested in). Screenshot taken directly from the `diatoms.xlsx` file.

Find the working directory

Use `getwd()` to check the current working directory. This is the location on your computer where R will look for files.

```
CODE
getwd()
```

- If you are working with a document file (like a Quarto or R Markdown file), the working directory is the location of the document file.
- If you are working with a script file (like an R script), the working directory might be different! You can use `setwd()` to change the working directory, **but this is not recommended**¹. Instead, use RStudio projects to set the working directory — see Tutorial 1 for more information or check out the [documentation](#) from Posit.

Move the file

Move `diatoms.xlsx` to the working directory. This will simplify the file path for data import.

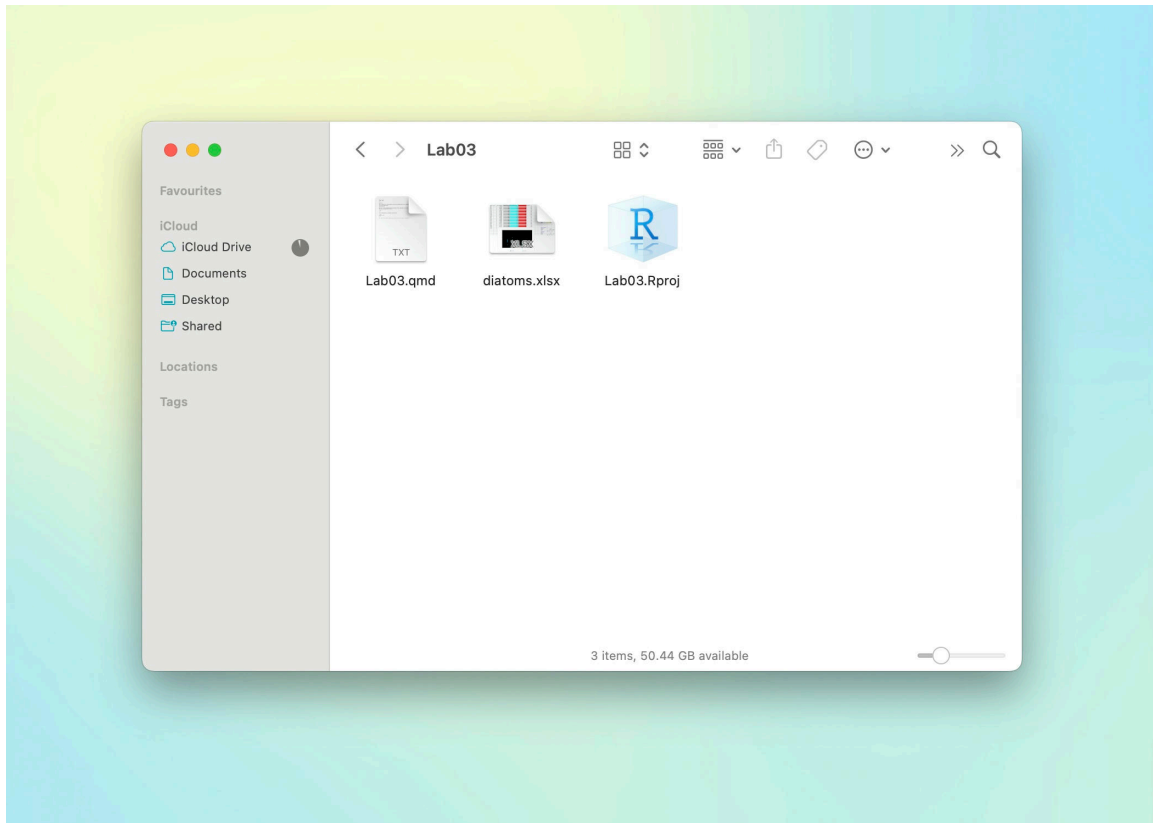


Figure 2: Place data in the working directory to simplify the file path for data import. In this screenshot, `diatoms.xlsx` has been placed in the folder that contains an `.Rproj` file.

Then, use `read_excel()` from the `readxl` package to read the file, noting the file path, the worksheet name, and the range of cells that contain the data. Below we have assigned the output to an object called `diatoms`.

```
CODE
library(readxl)
diatoms <- read_excel("data/diatoms.xlsx",
  sheet = "Diatoms",
  range = "A1:D35"
)
```

¹The `setwd()` function is not recommended because it is not reproducible: it can cause issues when sharing your code with others, or when working on different computers. Instead, use RStudio projects to set the working directory.

Moving the file and updating the path

Suppose you want to organise your project files and put all data files in a single folder. You can do this by creating a new folder in the same directory as your Quarto file, and moving the data file(s) into it.

Here, we create a folder called data and move the `diatoms.xlsx` file into it.

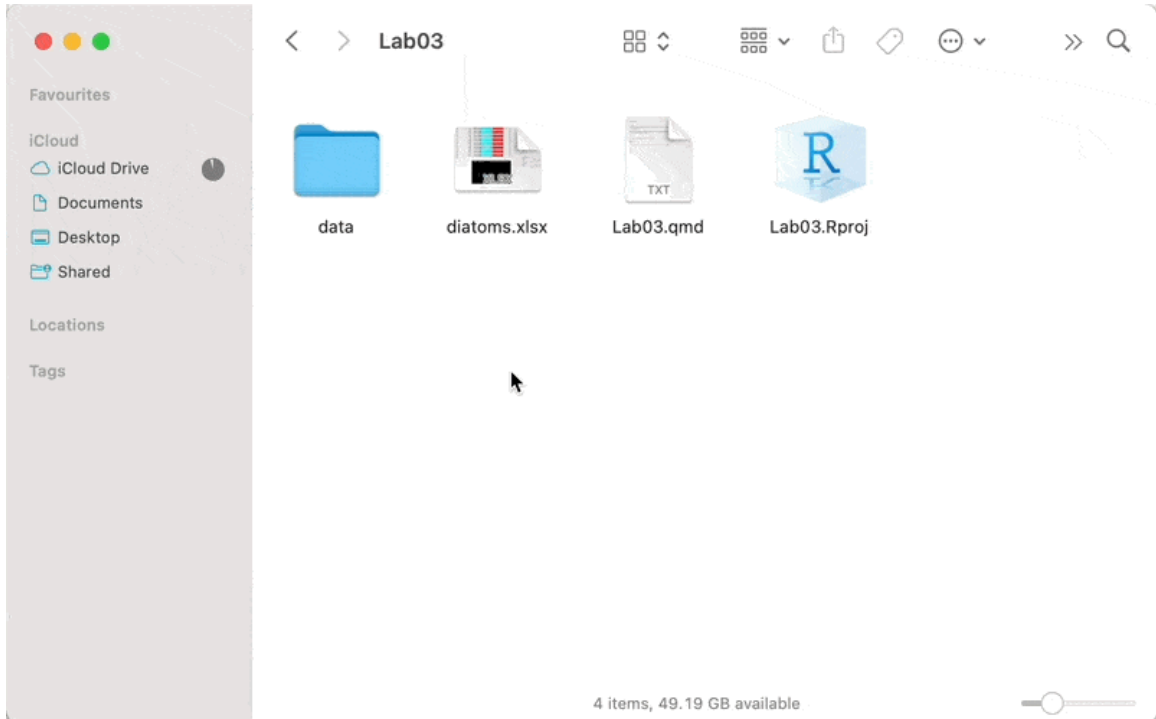


Figure 3: Create a folder called data (but it can be any other name, really), and move your data file(s) to this folder. You will have to update the path to the file in the `read_excel()` function to read the data from the new location.

We then update the path to the file in the `read_excel()` function by adding the folder name `data/` to the file path.

```
CODE
diatoms <- read_excel(
  path = "data/diatoms.xlsx",
  sheet = "Diatoms",
  range = "A1:D35"
)
```

Exercises

Here are some quick exercises to see if you can navigate your way around file paths in R.

1. Create a new folder called `output` inside the `data` folder and move the `diatoms.xlsx` file into it. How will you update the path in the `read_excel()` function to reflect this change?
2. Rename the `output` folder to `raw`.
3. Rename the `data` folder to `diatoms` and update the code.